Training for students using cryogenic material (LN2) in the stage 3 and 4 laboratories

These items are taken from the risk assessment UCDC4 Handling and use of Cryogenic Liquids (general). This training has been designed for the purpose of undergraduate experiments that use LN2 stored in 1-litre flasks that have been filled by a staff member. Some topics have been removed that are not relevant to student use such as items that deal with handling, use, storage or transport of large volumes.

Topics	**Covered in training**
1. No person may handle or use cryogenic material without being suitably trained in its safe use. It is the responsibility of laboratory supervisors / managers to ensure that all persons under their control using cryogens have been trained, and that full records of such	
2. Safety Data Sheets for each cryogenic material must be available for any such material in use in the laboratory.	
3. Suitable Personal Protective equipment must be worn when handing or working with cryogenic liquids. At a minimum this should include: a. Face shield for handling large volumes of material or when pouring cryogenic liquids or filling containers	
b. Safety glasses when working with small volumes	
c. Protective gloves conforming to BS EN 511 (Cold Protection). The gloves should be specifically designed for cryogenic handling with ribbed cuffs to prevent splashing into the glove or be loose fitting gauntlets that can easily be removed. The glove material should be rough to give good grip while handling and not increase the chance of spillage	
d. Suitable apron for handling large volumes of material or decanting material.	
e. Closed laboratory coat	
f. Safety boots when handling large volumes or large containers. Boots must be worn in such a way that there is no risk of spilled material getting inside the boot (i.e. they must be tucked under trouser legs)	
4. Long sleeves and trouser legs must be worn when handling cryogenic material. There must be minimal exposed skin	

5. All metallic jewellery should be removed when handling cryogenic materials as metal items will quickly spread the cold from any contact with the cryogenic material.	

Storage

Topics	**Covered in training**
1. Storage vessels must be suitably rated for the holding of cryogenic materials. Vessels holding cryogenic materials will be subjected to cold temperatures and increased internal pressure. Carbon steel, plastics and rubber becomes brittle and may fracture at low temperatures.	
2. Storage vessels for liquid oxygen must be kept clean to avoid contamination with materials that might become flammable in an oxygen enriched environment.	
3. Purpose designed non pressurised vacuum flasks must be used to store volumes of cryogenic liquids between 1-50 litres, i.e. Dewar Flasks. Dewar Flasks must have non sealed stoppers to allow boiling material to vent. Any glass component of a Dewar flask must be covered in tape to prevent shattering in the event of an explosion.	
4. Always make sure that containers of liquid nitrogen are suitably vented and unlikely to block due to ice formation.	
5. Small Dewar flasks may be carried by hand. All other cryogenic liquid containers must be carried in purpose-built trolleys	

Use

Topics	**Covered in training**
1. When pouring cryogenic liquids do so slowly and carefully to minimise splashing and rapid cooling of the receiving container.	
2. Always use thongs when placing or removing items from cryogenic liquids.	
3. When disposing of cryogenic materials do not pour them down the sink or allow them to vaporise into enclosed areas such as laboratories, fridges, freezers, cold rooms, etc. Cryogenic materials to be disposed of through vaporisation must be left in well ventilated areas, e.g. a fume hood.	
4. Pregnant females and asthmatic workers must seek medical approval prior to working with cryogenic materials.	
5. Low temperature damage to the insulation on electrical cables can lead to electrocution and equipment damage. Cryogen users must ensure that cables are not placed where they can be affected by cryogen spills.	
6. Equipment cooled outside by liquid nitrogen but open to air may allow liquid oxygen to form inside which can create a dangerous pressure rise on warming or an explosion or fire in contact with flammable or combustible material. Use liquid nitrogen to cool sealed or evacuated systems only.	

7. Beware of the formation of liquid oxygen in cold traps that are open to air or the increase of liquid oxygen content in a flask of liquid nitrogen that has been cold for a long period.	
8. Students should stand at the experiment when using the liquid nitrogen in order to avoid exposure in the event of a spillage onto their lap from the bench.	

Emergency Response

Topics	**Covered in training**
Emergency Response 1. In the event of a cold burn from a cryogenic material: a. Remove any restrictive clothing - but not any that is frozen to the tissue b. Flush the affected area with tepid water (not above 40oC) to return tissue to normal body temperature c. Do not apply any direct heat or rub affected area d. Cover with a loose, sterile dressing and keep patient warm e. Obtain medical assistance immediately 2. All users of cryogenic material should be aware of the symptoms of anoxia (physiological oxygen depletion). These include dizziness, a narcotic type affect, nausea, confusion, etc. Persons experiencing such symptoms should remove themselves to fresh air. Persons observing such symptoms in co-workers should remove them to fresh air. In the event that breathing stops inform the local first aider and give artificial respiration. 3. Do not attempt to rescue anyone from a confined space if they were working with cryogenic materials and have lost consciousness - open the door and raise the alarm by contacting the UCD 24-hour Emergency Line on internal extension **7999** or **01 716 7999** from a mobile. 4. For minor spillages (<1 litre) of cryogenic material the following protocol should be followed: a. Shut off all sources of ignition if spill involves liquid oxygen or flammable materials if safe to do so b. Allow liquid to evaporate, ensuring adequate ventilation c. Following return to room temperature inspect area where spillage has occurred d. If there is any damage to the floors, benches or walls inform the buildings office on internal telephone **1111** e. If any laboratory equipment has been damaged following the spillage inform the laboratory manager / supervisor	

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